

Excel Assignment 1 – Data Exploration

Product Dataset: Data Summarization, Logical Functions, Conditional Aggregation & Text Extraction

Project / Work Description

This project applies core Excel data-exploration techniques to a 30-row product dataset containing Product ID, Product Name, Brand Name, Quantity, Category, and Price. The workbook computes summary statistics (SUM, COUNT, AVERAGE, MIN, MAX), classifies each product into a Price Range using an IF condition (\$500 threshold), performs conditional aggregation with SUMIF (total price of Electronics) and COUNTIF (products priced under \$100), and extracts structured fields — Day, Month, and Country Code — from the Product ID using LEFT, MID, and RIGHT text functions. Every result is produced by a live Excel formula (no hardcoded values), so the workbook recalculates automatically if the underlying dataset changes.

Dataset	Product Dataset — 30 records, 6 base attributes
Product ID format	DD-MMM-CC (e.g. 05-JAN-US → Day=05, Month=JAN, Country=US)
Functions used	SUM, COUNT, AVERAGE, MIN, MAX, IF, SUMIF, COUNTIF, LEFT, MID, RIGHT
Deliverable file	Excel_Assignment1_Data_Exploration.xlsx

1. Dataset — Columns A–D (Screenshot)

Product ID	Product Name	Brand Name	Quantity
05-JAN-US	Wireless Mouse	Logitech	120
12-JAN-IN	Bluetooth Speaker	boAt	80
18-JAN-UK	Cotton T-Shirt	H&M	200
22-JAN-US	Office Chair	IKEA	15
29-JAN-CA	LED Desk Lamp	Philips	60
03-FEB-US	Laptop Stand	AmazonBasics	90
09-FEB-IN	Running Shoes	Puma	70
14-FEB-AU	Smart Watch	Noise	40
19-FEB-UK	Ceramic Mug Set	IKEA	150
25-FEB-US	4K Monitor	Samsung	25
02-MAR-CA	Wool Sweater	Zara	85
08-MAR-IN	Study Table	Godrej	20
15-MAR-US	Noise Cancelling Headphones	Sony	35
21-MAR-UK	Kids Toy Car	Hot Wheels	110
27-MAR-AU	Standing Desk	IKEA	12
02-APR-US	Gaming Keyboard	Razer	55
07-APR-IN	Yoga Mat	Decathlon	130
13-APR-CA	Leather Wallet	Fossil	65
19-APR-UK	Blender	Philips	45
24-APR-US	External SSD 1TB	SanDisk	70
30-APR-AU	Backpack	Wildcraft	95
05-MAY-IN	Air Fryer	Philips	30
11-MAY-US	Bookshelf	IKEA	18
16-MAY-UK	Wireless Earbuds	boAt	140
22-MAY-CA	Denim Jacket	Levi's	75
28-MAY-US	Smartphone	Samsung	22
03-JUN-IN	Office Desk	Godrej	10
09-JUN-UK	Board Game	Hasbro	85
14-JUN-AU	Tablet	Lenovo	28
20-JUN-US	Coffee Maker	Nescafe	40

1. Dataset — Columns E–I incl. Newly Created Columns (Screenshot)

Category	Price (\$)	Price Range	Day	Country Code
Electronics	\$25.99	Standard Price	05	US
Electronics	\$45.50	Standard Price	12	IN
Clothing	\$12.99	Standard Price	18	UK
Furniture	\$150.00	Standard Price	22	US
Electronics	\$32.75	Standard Price	29	CA
Electronics	\$28.40	Standard Price	03	US
Clothing	\$55.00	Standard Price	09	IN
Electronics	\$89.99	Standard Price	14	AU
Kitchen	\$18.25	Standard Price	19	UK
Electronics	\$310.00	Standard Price	25	US
Clothing	\$42.00	Standard Price	02	CA
Furniture	\$135.50	Standard Price	08	IN
Electronics	\$199.99	Standard Price	15	US
Toys	\$9.99	Standard Price	21	UK
Furniture	\$420.00	Standard Price	27	AU
Electronics	\$75.00	Standard Price	02	US
Sports	\$15.75	Standard Price	07	IN
Accessories	\$38.50	Standard Price	13	CA
Kitchen	\$52.30	Standard Price	19	UK
Electronics	\$95.00	Standard Price	24	US
Accessories	\$34.99	Standard Price	30	AU
Kitchen	\$110.00	Standard Price	05	IN
Furniture	\$89.00	Standard Price	11	US
Electronics	\$29.99	Standard Price	16	UK
Clothing	\$68.00	Standard Price	22	CA
Electronics	\$650.00	High Price	28	US
Furniture	\$275.00	Standard Price	03	IN
Toys	\$19.50	Standard Price	09	UK
Electronics	\$220.00	Standard Price	14	AU
Kitchen	\$64.00	Standard Price	20	US

Columns G–J (Price Range, Day, Country Code, Month) are newly created, formula-driven columns.

2. Summary Calculation Results (Screenshot)

Summary Calculations

Total Price of All Products	\$3,413.38
Number of Products	30
Average Price	\$113.78
Minimum Price	\$9.99
Maximum Price	\$650.00
Total Price - Electronics Category (SUMIF)	\$1,802.61
Count of Products Priced Below \$100 (COUNTIF)	21

3. Applied Formulas — Formula Bar (Screenshots)

Representative formula-bar captures for each function used in the workbook:

Total Price — SUM

D35	fx	=SUM(F2:F31)
-----	----	--------------

Number of Products — COUNT

D36	fx	=COUNT(F2:F31)
-----	----	----------------

Average Price — AVERAGE

D37	fx	=AVERAGE(F2:F31)
-----	----	------------------

Minimum Price — MIN

D38	fx	=MIN(F2:F31)
-----	----	--------------

Maximum Price — MAX

D39	fx	=MAX(F2:F31)
-----	----	--------------

Price Range classification — IF

G2	fx	=IF(F2>=500,"High Price","Standard Price")
----	----	--

Electronics category total — SUMIF

D40	fx	=SUMIF(E2:E31,"Electronics",F2:F31)
-----	----	-------------------------------------

Products under \$100 — COUNTIF

D41	fx	=COUNTIF(F2:F31,"<100")
-----	----	-------------------------

Day extraction — LEFT

H2	fx	=LEFT(A2,2)
----	----	-------------

Month extraction — MID

J2	fx	=MID(A2,4,3)
----	----	--------------

Country Code extraction — RIGHT

12	fx	=RIGHT(A2,2)
----	----	--------------

Prepared by: Dhanush — B.Tech AI & Data Science, Dhaanish Ahmed College of Engineering (Anna University)